

Maximum Likelihood Estimation for Binomial Observations

COM3031 Probabilistic Machine Learning

1 Problem Setup

Consider the following experiment:

- A coin is flipped n times.
- Each flip is independent.
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$$y_i = \begin{cases} 1, & \text{if the } i\text{-th flip is Heads,} \\ 0, & \text{if the } i\text{-th flip is Tails,} \end{cases} \quad i = 1, \dots, n.$$

Let $y = \sum_{i=1}^n y_i$ denote the total number of heads observed. We assume each flip follows a Bernoulli distribution with unknown parameter $\theta = \Pr(\text{Heads})$, $0 \leq \theta \leq 1$.

Our goal is to estimate θ using Maximum Likelihood Estimation (MLE).

2 Likelihood Function

The number of heads in n independent trials follows a binomial distribution:

$$p(y \mid \theta) = \binom{n}{y} \theta^y (1 - \theta)^{n-y}.$$

3 Log-Likelihood

Taking the logarithm of the likelihood gives:

$$\ell(\theta) = \log p(y \mid \theta) = \log \binom{n}{y} + y \log \theta + (n - y) \log(1 - \theta).$$

The term $\log \binom{n}{y}$ does not depend on θ and can be treated as a constant during optimisation.

4 Optimisation

To find the maximum likelihood estimate, we differentiate the log-likelihood with respect to θ and set the derivative equal to zero:

$$\frac{d\ell(\theta)}{d\theta} = 0.$$

Computing the derivative:

$$\frac{d\ell(\theta)}{d\theta} = \frac{y}{\theta} - \frac{n - y}{1 - \theta}.$$

5 Stationary Condition

Setting the derivative equal to zero:

$$\frac{y}{\theta} - \frac{n-y}{1-\theta} = 0.$$

Rearranging:

$$\frac{y}{\theta} = \frac{n-y}{1-\theta}.$$

Cross-multiplying:

$$y(1-\theta) = (n-y)\theta.$$

Expanding and simplifying:

$$y - y\theta = n\theta - y\theta, \quad \Rightarrow \quad y = n\theta.$$

6 Maximum Likelihood Estimate

Solving for θ gives:

$$\boxed{\hat{\theta}_{\text{MLE}} = \frac{y}{n}}$$

7 Interpretation

For binomial observations, the maximum likelihood estimate of the success probability is the sample proportion:

$$\hat{\theta}_{\text{MLE}} = \frac{\text{number of successes}}{\text{number of trials}}.$$